

Unsupervised Concept Discovery for Medical Vision–Language Models:

A Rigorous Characterization of Sparse-Autoencoder Failure and Deterministic Alternatives

Marc’Antonio Lopez
Politecnico di Torino
Turin, Italy

Nicolò Colle
Politecnico di Torino
Turin, Italy

Carmine Francesco Benvenuto
Politecnico di Torino
Turin, Italy

ABSTRACT

Medical Vision–Language Models (VLMs) are accurate but opaque, and concept-based explainability promises to expose their internal knowledge as human-readable units. We re-implement the MedConcept pipeline [5]—sparse autoencoder (SAE) decomposition of a frozen VLM’s image embeddings, alignment to a curated radiology vocabulary, and LLM-as-judge evaluation—on BiomedCLIP [14], on two scales: IU X-Ray [3] (~7k images) and ROCov2 (~80k, a 10× scale test). Our first result is methodological: the slot-wise cross-seed Jaccard used as the stability headline is *floor-by-construction*—a single SAE whose features are relabelled by a random bijection scores ≈ 0.0077 , indistinguishable from real cross-seed—so it cannot establish non-identifiability. We therefore report a permutation-invariant *matched* metric against a subspace-conditioned (effective-rank) null. Under this honest metric, SAE concepts are genuinely non-identifiable: best-match cosine 0.30–0.33 at only $1.67\text{--}1.81\times$ the conditioned null, with $\approx 0\%$ of features matching at ≥ 0.9 across all seed pairs, and naming cosine (mean) of 0.40–0.48 (0% above 0.7) even with a chest-specific vocabulary. Crucially, the 10× ROCov2 scale does *not* lift identifiability (matched ratio $1.67\times \rightarrow 1.81\times$, $\text{frac} \geq 0.9 \text{ } 0\% \rightarrow 0.3\%$), refuting the data-scale hypothesis and localising the binding constraint to the *representation site*: SAE-on-CLS decomposes an already-compressed, contrastive-shaped global vector, and no dictionary over 30–500× the input’s intrinsic dimension is identifiable. SPLiCE [1], a deterministic decomposition over a fixed dictionary, sidesteps the estimation problem and is the sole judge-ready method. We report these outcomes honestly—including the slot-wise metric artefact—as a reproducibility study of *where* unsupervised concept discovery breaks and *why*.

KEYWORDS

Concept-based explainability, sparse autoencoders, medical vision–language models, interpretability evaluation, reproducibility

1 INTRODUCTION

Medical VLMs attain strong performance on pathology classification, segmentation, and report generation, yet their internal representations are high-dimensional, polysemic, and opaque—a serious concern in safety-critical clinical use. Classical post-hoc tools such as saliency maps and attention visualisations produce local explanations that seldom reveal the semantic structure the model has actually learned.

Concept-based explainability reframes an explanation in terms of human-interpretable units. Where supervised approaches such as Concept Bottleneck Models [7] and TCAV [6] require a pre-defined,

labelled concept set, recent work attempts *unsupervised* discovery: decomposing a frozen model’s representations into sparse features, naming each feature against a clinical vocabulary, and validating the result quantitatively. MedConcept [5] instantiates this as a three-stage pipeline (sparse-activation extraction, vocabulary alignment, LLM-judge evaluation) and introduces the Aligned/Unaligned/Uncertain metrics for quantitative interpretability assessment.

A central but under-reported question is whether the concepts such pipelines discover are *reproducible* and *clinically valid*. In this work we re-implement the MedConcept paradigm faithfully, and find—rigorously, and against analytical nulls—that the learned sparse decomposition is essentially non-identifiable: independently seeded SAEs share no canonical features (a permutation-invariant matched metric lands at $1.67\text{--}1.81\times$ a subspace-conditioned null, with $\approx 0\%$ of features matching at ≥ 0.9 cosine), the naming signal is capped at 0.40–0.48 cosine, and a large fraction of features never fire. Crucially, we first expose that the slot-wise index Jaccard used as the stability headline is *floor-by-construction* (a relabelled copy of one SAE scores ≈ 0.0077 on it), so non-identifiability must be measured permutation-invariantly.

We then investigate three directions intended to address the root causes, and an organisational extension:

- **Baseline** (512-d projected SAE): a faithful re-implementation of MedConcept, characterised as a documented failure case.
- **Path A** (768-d pre-projection hidden state SAE): targets the representation-location mismatch hypothesised to drive non-identifiability.
- **Path B** (SPLiCE): replaces the learned dictionary with a fixed RadLex dictionary, removing the estimation problem entirely.
- **Extension** (concept organisation): clusters flat concept lists into structured, hierarchy-grounded families.

Contributions. (1) A *relabeling control* proving the slot-wise cross-seed Jaccard is floor-by-construction: one SAE with its features renamed by a random bijection scores ≈ 0.0077 on it, yet 1.0 on a permutation-invariant matched metric—so the slot-wise number measures feature *indexing*, not feature *identity*, and cannot establish non-identifiability. (2) A permutation-invariant *matched-stability* metric with a *subspace-conditioned* (effective-rank) null that controls for data-manifold concentration; under it, SAE concepts are genuinely non-identifiable (best-match $1.67\text{--}1.81\times$ the null, $\approx 0\%$ at ≥ 0.9). (3) A direct test of the data-scale hypothesis: ROCov2 [12] at 10× IU X-Ray scale does *not* restore identifiability, localising the binding constraint to the representation site (SAE-on-CLS), not data

volume. (4) An honest, partially-negative evaluation—including the slot-wise artefact—rather than a curated success story.

2 RELATED WORK

Concept-based explainability. Concept Bottleneck Models [7] restructure a network to predict an explicit concept layer before the classifier, trading minor accuracy for auditable, concept-level intervention. TCAV [6] tests directional sensitivity of a logit to user-defined Concept Activation Vectors. Both presuppose supervised, pre-defined concepts—precisely the limitation unsupervised discovery targets.

Sparse autoencoders for monosemantic decomposition. Bricken et al. [2] showed that overcomplete dictionary-learning autoencoders trained on transformer activations yield individual latents that activate on coherent, human-nameable features. Gao et al. [4] scaled the Top-K (k-sparse) autoencoder, which sets the L_0 sparsity exactly via top- k gating and removes the L1 sparsity penalty, improving the reconstruction-sparsity frontier; we adopt this architecture.

SAE stability and universality. A growing line asks whether SAE features are canonical. Lan et al. [8] showed that SAE feature spaces across language models are similar only up to rotation, sharing a subspace rather than identical features. Leask et al. [10] argued SAEs are simultaneously incomplete and non-atomic, and therefore “do not find canonical units of analysis.” This weak-vs-strong universality distinction motivates our matched-stability metric: we no longer expect identical features across seeds, only a shared subspace.

Sparse concept decomposition. SPLiCE [1] bypasses a learned dictionary and decomposes a frozen CLIP embedding directly into a sparse, non-negative combination over a fixed, interpretable concept dictionary. Because the dictionary is fixed and the method deterministic, it sidesteps the non-identifiability of learned factorisation—a property we exploit as a data-frugal baseline.

Medical VLMs and the reference framework. BiomedCLIP [14] is a domain-specific contrastive VLM (ViT-B/16 image encoder + PubMedBERT text encoder, pretrained on PMC-15M) whose image encoder exposes a 768-d pre-projection hidden state; its released checkpoint projects this into the 512-d shared contrastive space our pipeline decomposes. MedConcept [5] provides the methodological template—sparse extraction, vocabulary alignment, and LLM-judge evaluation with the Aligned/Unaligned/Uncertain metrics—which we re-implement and stress-test. RadLex [9] provides the curated radiology lexicon used for naming and organisation.

3 RESEARCH GAPS

From the literature and the project brief we identify six gaps that this work addresses:

- (1) **Concept instability / non-identifiability.** Learned sparse factorisations are not guaranteed to be reproducible across seeds, and in the small-data regime the decomposition is provably non-identifiable. No prior medical-VLM concept work we are aware of reports seed-stability against an analytical null.
- (2) **Clinical validity of discovered concepts.** A cosine-assigned name is cosmetic unless it tracks real pathology; concepts are rarely validated against held-out ground-truth labels.
- (3) **Representation-location mismatch.** The SAE-interpretability literature operates on raw hidden states, yet reference pipelines fit the SAE on the already-projected, modality-gap-bearing CLIP space—a structurally poorer target.
- (4) **Small-data robustness vs. learned dictionaries.** Learned SAEs require 10^5 – 10^6 activations; medical corpora are 2–3 orders of magnitude smaller, yielding large dead-feature fractions.
- (5) **Concepts treated as independent.** Concept explanations are typically flat top- k lists, ignoring natural anatomical/hierarchical structure.
- (6) **Non-scalable, non-reproducible evaluation.** Interpretability claims typically rest on qualitative inspection or single-metric summaries whose chance floor is never reported—e.g. slot-wise feature-index overlap is degenerate under the arbitrary per-seed feature permutation—so claims are neither reproducible nor falsifiable. A quantitative metric must be calibrated against an analytical null and be permutation-invariant.

4 METHODOLOGY

4.1 Backbone, vocabulary, and split

We use BiomedCLIP (ViT-B/16 image encoder with a 768-d hidden state, projected by a frozen visual_projection : $\mathbb{R}^{768} \rightarrow \mathbb{R}^{512}$ into a shared contrastive space). Images are resized to 224×224; both image and text embeddings are L2-normalised at extraction. The concept vocabulary is built from RadLex: terms are encoded with BiomedCLIP’s text encoder, ranked by cosine similarity to a multi-centroid relevance score computed from 39 anchor queries across 13 clinical sub-domains, and the top terms plus 14 NIH ChestX-ray14 seed terms are retained, yielding ~1,030 chest-radiography-relevant terms cached as 512-d text embeddings.

The shared 512-d space exhibits a *modality gap* [11]: a systematic offset between the image and text cones (cross-modal image–text cosine ≈ 0.27 , centroid L2 distance ≈ 0.94). Following prior work we estimate it as the difference of the mean image and vocabulary embeddings and subtract it before cosine naming/decomposition.

All experiments use *two* chest-radiography corpora. IU X-Ray [3] (7,470 frontal/lateral images across 3,852 radiographic studies) is the primary, small-scale setting; ROCov2 (~80k images) is a 10×-scale replication that lets us test the data-volume hypothesis directly. In order to avoid data leakage, we split IU X-Ray *by radiographic study* (IU key = patient_IM-study), so no IU study appears at the same time in both train and test sets; in opposite, ROCov2 is simply split by image, since there is no shared patient structure to leak. The IU split yields 5,955 train / 1,515 test images (3,081 / 771 studies) with verified group overlap = 0, recomputed deterministically on every run (sorted study set, fixed seed).

4.2 Baseline: Top-K SAE on the 512-d projected space

We train a Top-K SAE [4] ($\text{ReLU}(W_{\text{enc}}(x - b_{\text{dec}}) + b_{\text{enc}})$), hard top- k sparsification, b_{dec} initialised to the geometric median of training

data) on the 512-d image embeddings. Configuration: dictionary size $D=2048$, $k=32$, 8,000 steps, $\text{lr}=5\times 10^{-5}$, batch size 256, five seeds (0, 42, 123, 456, 789). Concept naming assigns each *live* (non-dead) decoder row to its closest RadLex term by cosine similarity, after gap correction ($W_{\text{dec}} \leftarrow W_{\text{dec}} - \text{gap}$). Per-image pseudo-reports list the top-5 active concepts with a stated dominant concept.

4.3 Path A: Top-K SAE on the 768-d hidden state

Path A addresses Gap 3. Instead of `visual_projection(·)`, we extract the 768-d CLS token *before* the projection and train an identical Top-K SAE on these raw (un-normalised) hidden states. Because RadLex text lives in 512-d, we bridge the two spaces with the *frozen* projection: $\hat{W}_{\text{dec}}^{512} = W_{\text{dec}}^{768} W_{\text{proj}}^T$, gap-correct $\hat{W}_{\text{dec}}^{512}$, then name against RadLex as in the baseline. Dead features are detected pre-projection (a zero 768-d row projects to $-\text{gap}$ with non-zero norm) and cross-checked with an activation-based mask.

4.4 Path B: SPLiCE on the RadLex dictionary

Path B addresses Gaps 1 and 4. We decompose each gap-corrected 512-d image embedding into a sparse, non-negative combination over the fixed RadLex text dictionary with a two-stage solver (SPLiCE-style, in place of its original Lasso): Orthogonal Matching Pursuit selects $k=32$ atoms, then non-negative least-squares re-solves the coefficients on that support. The dictionary is never estimated from the $\sim 6\text{k}$ samples, so the method is deterministic by construction and has no cross-seed instability. The output schema mirrors the SAE explanations so the same judge can score it.

4.5 Concept-organisation extension

Addressing Gap 5, we cluster the active vocabulary into concept families by agglomerative clustering (average linkage, cosine metric). Each cluster is annotated with the most specific shared RadLex anatomical ancestor, with two degeneracy guards: leaf-root rejection and a *subtree-size canopy* that rejects candidate ancestors whose descendant count exceeds 1% of the ontology (e.g. the trivially generic “anatomical entity” node), so labels are honest rather than vacuously generic. The stage emits structured per-image families plus organisation metrics (silhouette, redundancy reduction, RadLex coverage). It is standalone and does not feed the judge.

4.6 Evaluation metrics

Stability (matched, the headline). For each decoder row in seed i we take the best cosine match across all rows of seed j (decoder-cosine matching [2, 8]), a permutation-invariant statistic. The null is a *subspace-conditioned* random-vector draw: real decoder directions concentrate in a subspace of effective rank $\text{erank}=(\Sigma\sigma)^2/\Sigma\sigma^2\approx 357-363$ (not the ambient 512), so we draw null vectors inside each decoder’s top- erank right-singular subspace and compare against the other decoder projected into it. This controls for the data-manifold concentration that an isotropic (full-512-d) null misses; the isotropic ratio is a reported lower bound. **Stability (slot-wise, deprecated).** Cross-seed Jaccard of top- k index sets has analytical floor $k/(2D-k)$, but is *floor-by-construction*: it is not invariant to the arbitrary per-seed feature permutation, so a single SAE whose features are relabelled by a random bijection scores ≈ 0.0077 on it (Sec. 5.2). We retain it only as an identical-permutation sanity

Table 1: Baseline Top-K SAE ($D=2048$, $k=32$) on IU X-Ray ($\sim 6\text{k}$) and ROCov2 ($\sim 80\text{k}$, $10\times$). Matched-cosine is permutation-invariant; “sub./iso. null” ratios are observed best-match over the subspace-conditioned and isotropic nulls.

	IU X-Ray (1 $\times$)	ROCov2 (10 $\times$)
Reconstruction cosine	0.991	0.968
Dead features (full test)	15.0–17.8%	0.5–0.8%
Matched cosine (obs)	0.299	0.327
Decoder erank	363	357
Subspace null ratio (honest)	1.67 $\times$	1.81 $\times$
Isotropic null ratio (lower bnd)	1.98 $\times$	2.16 $\times$
Frac. matched ≥ 0.9	0.0%	0.3%
Naming cosine (live, mean)	0.40	0.48
Slot-wise Jaccard (deprecated)	0.0077	0.0077

check and report it explicitly deprecated. **Naming.** Mean cosine of the assigned term; we additionally report the fraction of features matched at ≥ 0.7 and ≥ 0.9 , the concentration-robust signal (random directions cannot reach 0.9). **Faithfulness.** Point-biserial correlation of feature activations with in-distribution IU X-Ray MeSH/Problems labels, with a per-feature shuffle null. **LLM judge.** Each concept is scored Aligned/Unaligned/Uncertain against the reference report by a prompted LLM; faithfulness is reported as a *LIFT* (method %Aligned over a count-matched null’s %Aligned) to control for methods emitting different concept counts.

5 EXPERIMENTS AND ANALYSIS

5.1 The baseline is a non-identifiable failure case (Gap 1)

Table 1 reports the parity baseline ($D=2048$, $k=32$, 8k steps, five seeds) on both corpora. Reconstruction is faithful (cosine 0.968–0.991) yet cross-seed feature identity is absent: the permutation-invariant matched cosine is 0.30–0.33, only 1.67–1.81 $\times$ the subspace-conditioned null, and the fraction of features matched at ≥ 0.9 is $\leq 0.3\%$. Naming cosine is capped at 0.40–0.48 with 0% of features above 0.7—the closest RadLex term to a live decoder row scores only ≈ 0.40 cosine. We read this as genuine non-identifiability of the learned factorisation, not a training bug (init distinct, decoders full-rank, no collapse). The deprecated slot-wise Jaccard lands at ≈ 0.0077 , equal to its analytical floor—as expected, since it is floor-by-construction (Sec. 5.2).

5.2 Relabeling control: the slot-wise metric is floor-by-construction

The slot-wise index Jaccard compares feature *indices* across two SAEs with no alignment of which feature is which. Since independently seeded SAEs have no canonical feature ordering, the metric cannot distinguish “two genuinely different SAEs” from “one SAE whose columns have been renamed.” We verify this directly: we take one trained ROCov2 SAE, permute its encoder/decoder rows by a random bijection π (a function-preserving relabelling—reconstruction cosine identical to 10^{-4}), and measure both metrics between the original and the relabelled copy. Slot-wise Jaccard collapses to 0.0077 (the chance floor, indistinguishable from real

Table 2: Baseline (512-d) vs. Path A (768-d), parity configuration. Matched-cosine is permutation-invariant; “obs/null” is observed best-match cosine over the isotropic null.

	Baseline 512-d	Path A 768-d
Slot-wise Jaccard	0.0084	0.0083
Analytical floor	0.0079	0.0079
Dead features (full test)	16.0%	1.7%
Naming cosine (live)	0.42	0.47
Matched cosine (obs)	0.299	0.325
Matched cosine (null)	0.151	0.124
obs/null	1.97×	2.63×
Frac. matched ≥ 0.9	$\approx 0\%$	$\approx 0\%$

cross-seed 0.0077), while the matched metric correctly reports identity (1.0000, $\text{frac} \geq 0.9 = 1.0$). This isolates permutation noise as the entire slot-wise signal: it is a binary identical-permutation check, not a stability/identifiability metric. All subsequent stability claims use the matched metric.

5.3 Neither representation depth nor data scale restores identifiability

Table 2 contrasts Path A (768-d pre-projection hidden state) with the baseline. Operating on the 768-d hidden state markedly reduces dead features (1.7% vs. 16%) and improves naming cosine (0.47 vs. 0.42). Yet the permutation-invariant matched metric is unchanged in its verdict: both methods share a decoder subspace above the null (isotropic $1.97 \times / 2.63 \times$, collapsing to ≈ 1.3 – $1.5 \times$ under the subspace-conditioned null), and almost no features match strongly (0% at ≥ 0.9). This is *weak universality* [8, 10]: a shared subspace exists, but not canonical, reproducible features. Crucially, Path A still operates on the *pooled CLS token* (pre-projection, but still the contrastive-shaped global summary), so the representation-site mismatch is only partially addressed.

We then test the leading alternative explanation—data scale (Gap 4)—directly. Training the parity baseline on ROCov2 ($\sim 80k$ images, a $10 \times$ increase over IU X-Ray) improves training health (dead features $16\% \rightarrow 0.6\%$) but leaves identifiability flat (Table 1): matched cosine $0.299 \rightarrow 0.327$, subspace ratio $1.67 \times \rightarrow 1.81 \times$, $\text{frac} \geq 0.9$ $0.0\% \rightarrow 0.3\%$, naming capped at 0.40–0.48. A $10 \times$ corpus does not lift a single one of these above the non-identifiability regime. We conclude the binding constraint is *not* data volume but the *representation site*: SAE-on-CLS factorises a single, already-compressed, contrastive-shaped 512-d vector whose intrinsic dimension ($\text{erank} \approx 357$ – 363 , participation rank measured at 3.8–56) is far below the 2048-way dictionary, so infinitely many equal-reconstruction tilings exist and no feature is canonical.

A dictionary/ k sweep on Path A (Table 3) confirms the pattern: a more conservative ($D=1024$, $k=16$) configuration shows the strongest subspace signal (matched obs/null $2.78 \times$) and lowest naming noise, while aggressive overcompleteness ($D=4096$, $k=64$) dilutes it ($2.00 \times$)—but no setting escapes weak universality.

Table 3: Path A dictionary/ k sweep (768-d, 8k steps).

Preset	D	k	Dead	Matched obs/null
Conservative	1024	16	41.7%	$2.78 \times$
Default	2048	32	12.9%	$2.63 \times$
Aggressive	4096	64	6.6%	$2.00 \times$

5.4 Path B is deterministic and judge-ready

SPLiCE decomposes every test image with a fixed, randomness-free solver—Orthogonal Matching Pursuit selects the $k=32$ -atom support and non-negative least squares re-solves the coefficients on it—so the result is deterministic by construction and has no cross-seed instability. The output uses the SAE-compatible schema (feature_id/name/activation), so the same judge reads SPLiCE concepts directly. On IU X-Ray it covers all 1,515 test images in 9.5 s (CPU) using 997/1,031 vocabulary terms (96.7%), averaging 14.18 concepts/image (range 8–20); on ROCov2 ($\sim 80k$) it covers all 15,958 images in 101 s using 1,024/1,024 terms (100%), averaging 8.56 concepts/image (range 3–15). Modality-gap correction raises the maximum atom–image cosine sharply on *both* corpora (global maximum IU $0.49 \rightarrow 0.85$, ROCov2 $0.54 \rightarrow 0.86$; per-image mean $0.42 \rightarrow 0.79$ and $0.40 \rightarrow 0.75$), confirming the gap rather than the solver was the binding issue. The frequent terms are mixed rather than clean clinical findings: ROCov2 is dominated by imaging modalities (spiral CT, thoracic radiography, MR), reflecting its multi-anatomy corpus, while the IU vocabulary—a downloaded chest set—contains bilingual and off-topic entries (e.g. German translations such as *Umbilikaler venöser Katheter*, and non-chest terms such as *Backwash-Ileitis*); neither corpus yields the curated findings a hand-picked RadLex chest subset would, which is a property of the dictionaries supplied, not of SPLiCE itself.

5.5 Concept organisation is dense but weakly separated

Applied to all three IU sources (Table 4), the extension behaves as designed. SPLiCE, the densest source, groups 997 active concepts into 32 families, reducing concepts-per-image from 14.18 to 8.09 (redundancy reduction $1.75 \times$), but with weak cosine silhouette (0.020); because the supplied IU vocabulary is bilingual and off-topic (Sec. 5.4), every cluster receives a RadLex ancestor yet the labels inherit the dictionary’s noise (e.g. German *Lunge*, *Schmerz*, *Kolitis* alongside *vascular malformation*, *compression fracture*). The rebuilt SAE baseline is no longer silent—it exposes 80 active concepts (9 families, silhouette 0.094, zero empty images)—but these inherit the baseline’s non-identifiability: the families are real groupings of noisy names, not stable concepts. The Path A hidden-state source remains degenerate (14 active concepts, 1,260/1,515 empty images). Replicating the extension on ROCov2 SPLiCE gives 1,024 active concepts, 32 families, silhouette 0.066, zero empty images—non-degenerate, confirming density scales with corpus size, though cluster separation stays weak across both corpora.

Table 4: Concept organisation across sources.

Source	Active	Clusters	Silhouette	Empty img. (/1515)
SPLiCE	997	32	0.020	0
SAE baseline	80	9	0.094	0
SAE hidden	14	4	0.284	1260

5.6 A small upper tail of features is genuinely faithful

Despite global instability, a faithfulness ablation ($D=2048$ baseline, 5 seeds) finds that $17.8\% \pm 0.9$ of live features exceed the shuffle-null 95th percentile in point-biserial correlation with IU X-Ray MeSH/Problems labels, and 1.1% reach $|r| > 0.30$ (strongest $|r| = 0.459 \pm 0.057$, on a *mass* feature). This is a real but small upper-tail signal atop an otherwise degenerate dictionary: concepts are simultaneously *unstable cross-seed*—zero features cluster across the 5 seeds at the direction-identity threshold (pooled-decoder cosine ≥ 0.90), and only 0.1% of live rows sit in a cluster spanning ≥ 4 seeds at a relaxed cosine ≥ 0.70 , not significantly above a seed-tag shuffle-null ($p=0.17$)—and *partially faithful* (Gap 2).

5.7 The LLM-judge evaluation

The judge is fully implemented and its mechanics are correct and reproducible: a verb→label alias map (the prompt elicits SUPPORT/CONTRADICT/UNBINGUONS verbs while the schema wants Aligned/Unaligned/Uncertain, so verbs are normalised), temperature=0 with a global seed for determinism, infrastructure-error pairs excluded from the resume dedup so transient failures are retried, and the faithfulness metric is a count-matched *LIFT* (method %Aligned over a null emitting the same concept count). We do not report method-level LIFT rankings here: the run has not converged to completion, so absolute Aligned-rates are judge-dependent and only the within-judge LIFT against the count-matched null would be interpretable.

6 CONCLUSIONS

We set out to discover interpretable concepts in a medical VLM without supervision, and instead produced a rigorous characterisation of where and why this fails. Two methodological findings reframe the problem before any method comparison: the slot-wise cross-seed Jaccard used as the stability headline is *floor-by-construction* (a relabelled copy of one SAE scores ≈ 0.0077 on it, proved by control), so non-identifiability must be measured permutation-invariantly; and under the resulting matched metric with a subspace-conditioned null, SAE concepts are genuinely non-identifiable (best-match $1.67\text{--}1.81\times$ the null, $\approx 0\%$ at ≥ 0.9 , naming (mean) $0.40\text{--}0.48$). We then localised the cause: neither moving the SAE to the 768-d pre-projection hidden state (Path A) nor a $10\times$ increase in data (ROCOv2) lifts identifiability—the binding constraint is the *representation site*, not data volume. SAE-on-CLS factorises a single, already-compressed, contrastive-shaped global vector over a $30\text{--}500\times$ overcomplete dictionary, so no canonical features exist. SPLiCE, operating on a fixed dictionary, sidesteps the estimation problem and is the sole deterministic, judge-ready alternative. A concept-organisation extension still reduces redundancy in SPLiCE

output ($1.75\times$, 32 families), though cluster separation is weak (silhouette 0.020) and the family labels inherit the noise of the supplied dictionary, and a faithfulness ablation shows a small ($\sim 18\%$) but real upper tail of clinically-correlated features atop an otherwise degenerate dictionary.

Three lessons generalise beyond our setting. First, interpretability claims about discovered concepts must be calibrated against analytical nulls *and* a permutation-invariant statistic—slot-wise index overlap is degenerate and its floor can be mistaken for a result. Second, when a learned factorisation fails, data scale is the suspect of last resort: a $10\times$ corpus changed training health but not identifiability, because the failure was in the input representation. Third, an honest evaluation—including a metric artefact—is more scientifically useful than a curated one.

Limitations and future work. The LLM-judge evaluation is implemented (verb→label normalisation, temperature=0, count-matched LIFT) but not yet executed with MedGemma-4B [13] to convergence, so method-level rankings are deferred to future work with bootstrap confidence intervals. The decisive untested mitigation is the representation site itself: an SAE on the BiomedCLIP *patch-token residual stream* at a mid layer (not the pooled CLS), following the SAE-interpretability literature [2]; this is the experiment our diagnosis predicts would restore identifiability.

REFERENCES

- [1] Himabindu Lakkaraju, Suraj Srinivas, Flavio P. Calmon, and Himabindu Lakkaraju. 2024. Interpreting CLIP with Sparse Linear Concept Embeddings (SPLiCE). In *Advances in Neural Information Processing Systems (NeurIPS)*, Vol. 37. arXiv:2402.10376
- [2] Trent Bricken, Adam Templeton, Joshua Batson, Brian Chen, Adam Jermy, Nick Conder, Sushant Sellam, Christopher Olah, et al. 2023. Towards Monosemanticity: Decomposing Language Models with Dictionary Learning. Transformer Circuits Thread (Anthropic). <https://transformer-circuits.pub/2023/monosemantic-features/>
- [3] Dina Demner-Fushman, Marc D. Kohli, Marc B. Rosenman, Sonya E. Shooshan, Laritza Rodriguez, Sameer Antani, George R. Thoma, and Clement J. McDonald. 2016. Preparing a Collection of Radiology Examinations for Distribution and Retrieval. *Journal of the American Medical Informatics Association (JAMIA)* 23, 2 (2016), 304–310. <https://doi.org/10.1093/jamia/ocv080>
- [4] Leo Gao, Tom Dupré la Tour, Henk Tillman, Gabriel Goh, Rajan Troll, Alec Radford, Ilya Sutskever, Jan Leike, and Jeffrey Wu. 2024. Scaling and Evaluating Sparse Autoencoders. arXiv:2406.04093 [cs.LG]
- [5] Md Rakibul Haque, KM Arefeen Sultan, Tushar Kataria, and Shireen Elhabian. 2026. MedConcept: Unsupervised Concept Discovery for Interpretability in Medical VLMs. arXiv:2604.11868 [cs.CV]
- [6] Been Kim, Martin Wattenberg, Justin Gilmer, Carrie Cai, James Wexler, Fernanda Viegas, and Rory Sayres. 2018. Interpretability Beyond Feature Attribution: Quantitative Testing with Concept Activation Vectors (TCAV). In *Proceedings of the 35th International Conference on Machine Learning (ICML) (Proceedings of Machine Learning Research, Vol. 80)*. PMLR, 2668–2677. arXiv:1711.11279
- [7] Pang Wei W. Koh, Thao Nguyen, Yew Siang Tang, Stephen Mussmann, Emma Pierson, Been Kim, and Percy Liang. 2020. Concept Bottleneck Models. In *Proceedings of the 37th International Conference on Machine Learning (ICML) (Proceedings of Machine Learning Research, Vol. 119)*. PMLR, 5338–5348. arXiv:2007.04612
- [8] Michael Lan, Philip Torr, Austin Meek, Ashkan Khakzar, David Krueger, and Fazl Barez. 2024. Quantifying Feature Space Universality Across Large Language Models via Sparse Autoencoders. In *NeurIPS 2024 Workshop on Attributing Model Behavior at Scale*. arXiv:2410.06981
- [9] Curtis P. Langlotz. 2006. RadLex: A New Method for Indexing Online Educational Materials. *RadioGraphics* 26, 6 (2006), 1595–1597. <https://doi.org/10.1148/rg.266065168>
- [10] Patrick Leask, Bart Bussmann, Michael Pearce, Joseph Bloom, Curt Tigges, Noura Al Moubayed, Lee Sharkey, and Neel Nanda. 2025. Sparse Autoencoders Do Not Find Canonical Units of Analysis. In *International Conference on Learning Representations (ICLR)*. arXiv:2502.04878
- [11] Weixin Liang, Yuhui Zhang, Yongchan Kwon, Serena Yeung, and James Zou. 2022. Mind the Gap: Understanding the Modality Gap in Multi-modal Contrastive Representation Learning. In *Advances in Neural Information Processing Systems*

- (*NeurIPS*), Vol. 35. arXiv:2203.02053
- [12] Johannes Rückert, Louise Bloch, Raphael Brüngel, Ahmad Idrissi-Yaghir, Henning Schäfer, Cynthia S. Schmidt, Sven Koitka, Obioma Pelka, Asma Ben Abacha, Alba G. Seco de Herrera, Henning Müller, Peter A. Horn, Felix Nensa, and Christoph M. Friedrich. 2024. ROCov2: Radiology Objects in Context Version 2, an Updated Multimodal Image Dataset. *Scientific Data* 11 (2024). <https://doi.org/10.1038/s41597-024-03496-6> arXiv:2405.10004
- [13] Andrew Sellergren, Sahar Kazemzadeh, Tiam Jaroensri, et al. 2025. MedGemma Technical Report. arXiv:2507.05201 [cs.CV]
- [14] Sheng Zhang, Yanbo Xu, Naoto Usuyama, Hanwen Xu, Jaspreet Bagga, Robert Tinn, Sam Preston, Rajesh Rao, Mu Wei, Naveen Valluri, Cliff Wong, Andrea Tupini, Yu Wang, Matt Mazzola, Swadheen Shukla, Lars Liden, Jianfeng Gao, Angela Crabtree, Brian Piening, Carlo Bifulco, Matthew P. Lungren, Tristan Naumann, Sheng Wang, and Hoifung Poon. 2025. BiomedCLIP: A Multimodal Biomedical Foundation Model Pretrained from Fifteen Million Scientific Image-Text Pairs. arXiv:2303.00915 [cs.CV]